openLI

Update

Shane Alcock
NZNOG 2019

Previously…

- NZNOG 2018
  - Dave Mill proposed a project to develop software to solve TICSA LI
  - WAND would write the code
  - Operators would chip in money to cover development costs
  - Work together to test and deploy
  - Release finished code as open source

Sponsors

inspire.net
Local > Reliable > Fast > Broadband
Lightwire
VOCUS GROUP
catalyst
open source technologists
Trust power
NEW ZEALAND INTERNET EXCHANGE INC
NetSpeed
Rural Broadband
ispco.NZ
CCL

Mission Accomplished!

What is OpenLI?

- Runs on x86 Linux boxes
  - Packaged for Debian and Ubuntu

- Entirely passive
  - Mirror interceptable traffic into OpenLI collectors

- Relatively cheap
  - Software is OSS (GPLv3)
  - Collector hardware can be typical 1U servers + 10Gb NIC (for DPDK)

What OpenLI is NOT

- Effortless
  - Integration, security, testing, maintenance are still your problem
  - I've *tried* to make things as easy as possible

- A commercial-grade service
  - Support will be unofficial and best-effort

- Perfect (yet?)
  - There will be bugs, incompatibilities, etc.
  - Be nice to me and I'll be good to you ;)

We Did It!

- Core functionality is complete
  - IP Intercepts
  - RADIUS parsing to map IP sessions to users
  - VOIP Intercepts
  - Static IP ranges for IPv4 and IPv6
  - ETSI encoding of both IRIs and CCs
  - Mediation of encoded ETSI records to LEAs
  - Centralised provisioning
  - Distributed collection, including multiple interfaces per collector

© THE UNIVERSITY OF WAIKATO • TE WHARE WANANGA O WAIKATO

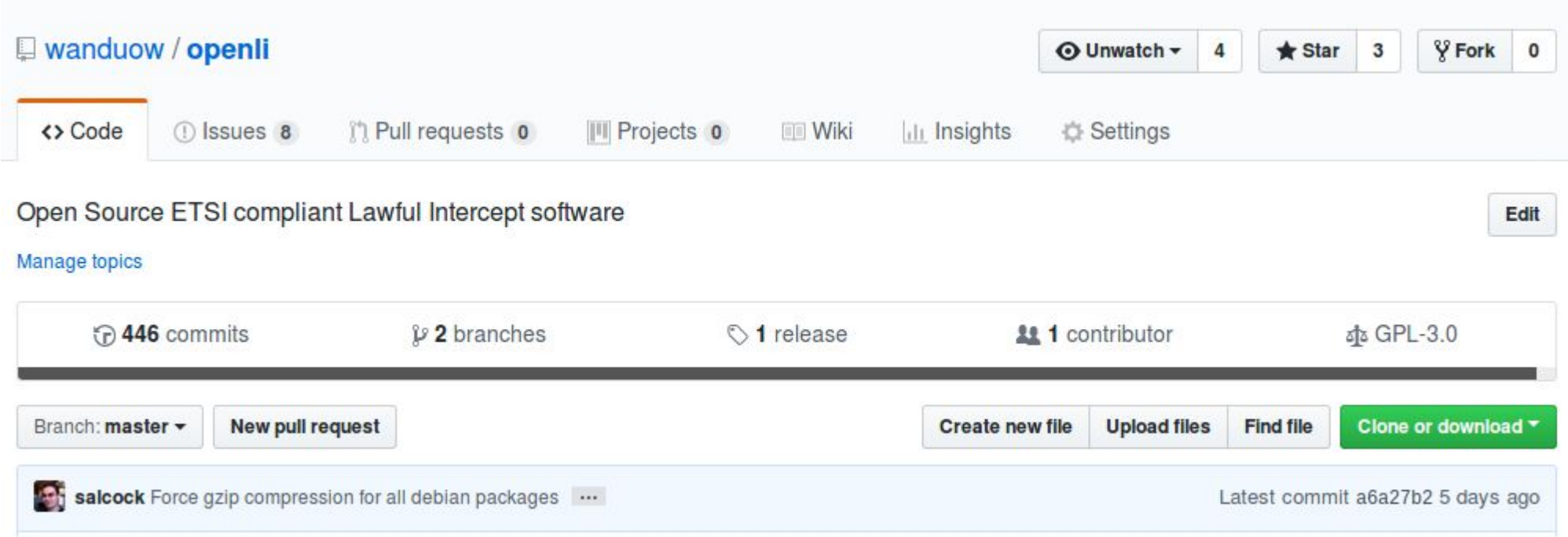
We Did It!

- Testing and deployment
  - At least one operator is running OpenLI in production
  - Tested with packet replays from other operators as well
  - Have also tested directly with NZ Police

© THE UNIVERSITY OF WAIKATO • TE WHARE WANANGA O WAIKATO

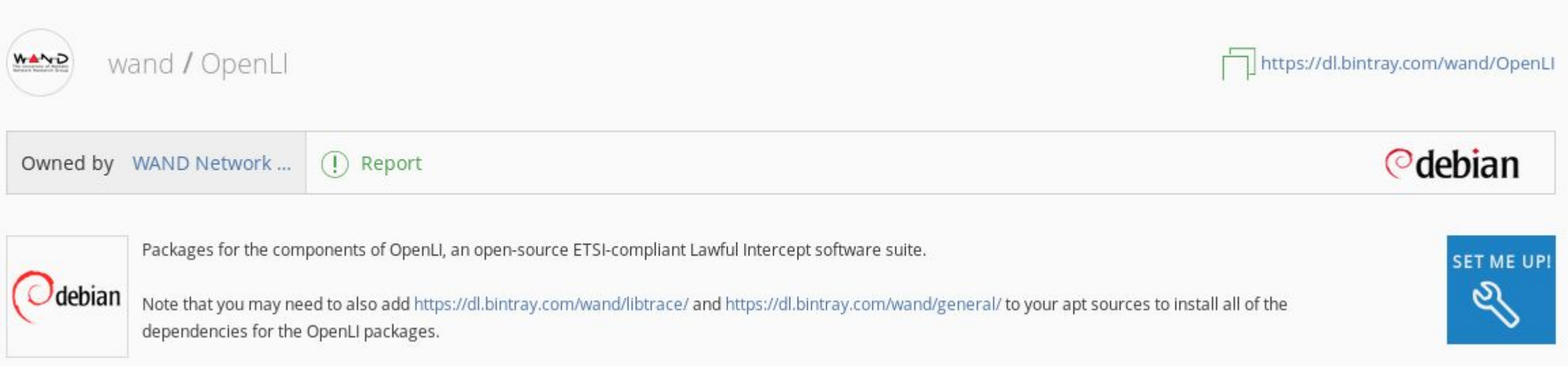
We Did It!

- Code has been released as open source
  - https://github.com/wanduow/openli
  - GPLv3 license



We Did It!

- Packaged for Debian / Ubuntu
  - https://bintray.com/wand/OpenLI/

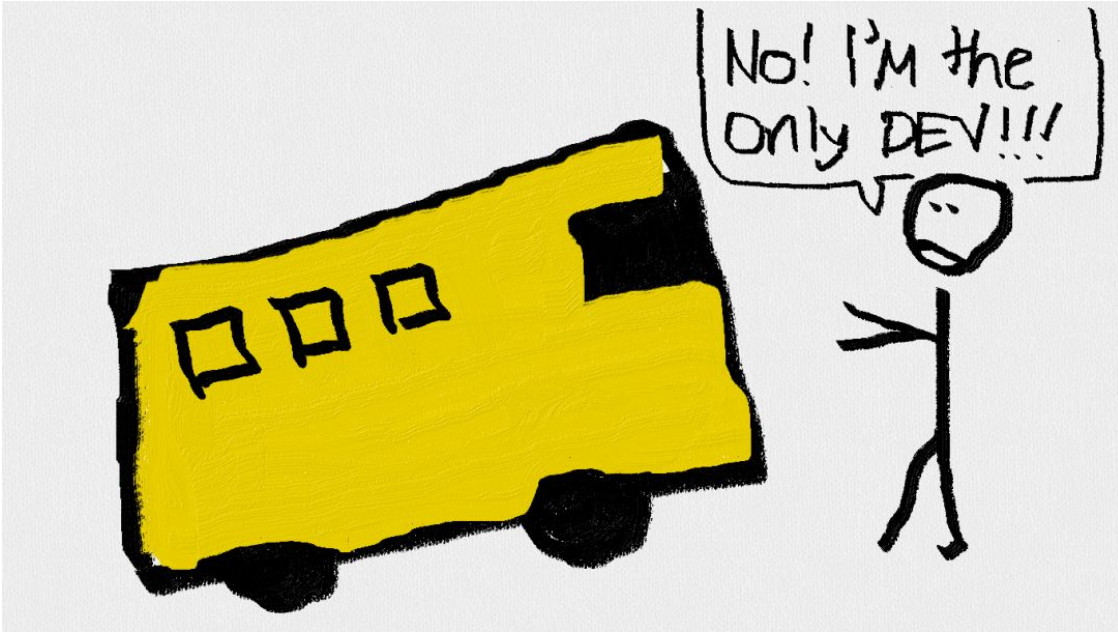


What’s Next?

- Use it!
  - Ran a tutorial here on Wednesday
  - If there is demand, we can arrange to run more tutorials

What's Next?

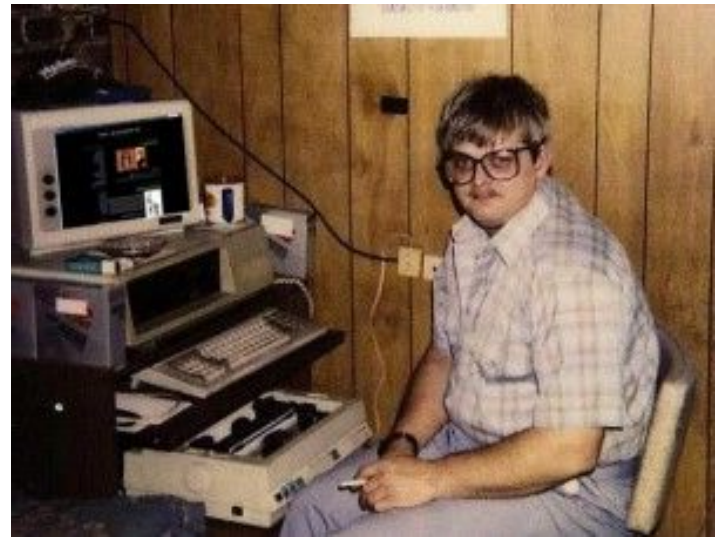
- Develop a user community around OpenLI
  - Bus factor is one (that's bad!)
  - Users can contribute examples as public documentation
  - Other developers adding features, fixing bugs



What's Next?

- Ongoing support
  - Each operator will have different expectations

Expectation



Reality

What's Next?

- "Best effort" support model
  - Deal with bug fixes and user feedback as my time permits
  - Prioritisation of support requests

- New development / features
  - Development fund account is empty
  - Want a feature added? Have spare cash? Talk to us!

© THE UNIVERSITY OF WAIKATO • TE WHARE WANANGA O WAIKATO

What's Next?

- Scope for future development
  - Live configuration updates via JSON objects
  - Disk-backed buffering to improve reliability
  - Strengthen security of internal communications
  - Translate vendor-specific LI into proper ETSI
  - Performance improvements

© THE UNIVERSITY OF WAIKATO • TE WHARE WANANGA O WAIKATO

Fin

Code: https://github.com/wanduow/openli

Website: https://openli.nz

Email us: contact@wand.net.nz